

CEPA Logic PRO v13 — Contextual Event Pattern Analysis

CEPA Logic PRO does not classify audio the way machine learning models do. Instead, it interprets patterns of behaviour over time, reacting to sequences of events rather than isolated blocks.

This document describes all contextual patterns used by AdBuster Hybrid Engine v13.

1. Trend Pattern (Slope Behaviour)

A trend is a consistent directional change across multiple audio blocks.

CEPA detects:

- upward trends (loudness gradually increases)
- downward trends (loudness gradually decreases)

Trends allow CEPA to anticipate events before they fully form.

2. Drift Pattern (Slow Deviation)

Drift is a slow, persistent shift away from the threshold.

Characteristics:

- gradual movement
- not a spike
- not an advertisement
- not a sudden change

CEPA applies gentle corrections to counter drift without overreacting.

3. Spike Pattern (Sudden Impulse)

A spike is a rapid, unnatural jump in RMS, DELTA or RANGE.

Typical sources:

- jingles
- transitions
- sudden noises
- short bursts of loud content

CEPA reacts with minimal, safe correction (maximum 1 step).

4. Stability Pattern (Background Consistency)

Stability means the signal stays within a stable zone for a period of time.

CEPA avoids corrections when:

- loudness is consistent
- no drift is forming
- no spikes occur

This prevents oscillation and keeps behaviour calm.

5. Deadzone Pattern (Human Tolerance)

Humans do not react to small fluctuations. CEPA mimics this behaviour using a deadzone:

- small changes are ignored
- jitter is eliminated
- behaviour stays natural

Deadzone = “no-reaction zone”.

6. Context Shift Pattern (Behavioural State Change)

CEPA transitions between behavioural states:

- NORMAL
- MUSIC
- TALK
- AD
- TRANSITION_UP
- TRANSITION_DOWN
- RETURN

Each context changes how CEPA interprets the same loudness value.

Examples:

- MUSIC → softer behaviour
- AD → aggressive protection
- TALK → dialogue margin

7. Return Pattern (Recovery to Baseline)

When loudness returns to normal, CEPA performs smooth, minimal corrections.

This prevents sudden jumps and keeps behaviour predictable.

8. Human-like Correction Pattern

CEPA spaces out actions to mimic human behaviour:

- cooldown between commands
- no repeated identical commands

- limited steps per time window
- anti-bounce protection
- anti-spam limits

This prevents oscillation and creates stable, human-like reactions.

9. Music Pattern (Soft Behaviour)

CEPA detects music using GUI dynamics and RMS behaviour.

Music mode:

- reduces sensitivity
- avoids reacting to transients
- protects musical dynamics

10. Dialogue Pattern (Talk Protection)

CEPA protects speech using the dialogue margin:

- avoids reacting to natural voice peaks
- prevents over-correction during conversations
- stabilizes spoken content

11. Advertisement Pattern (AD Mode)

AD Mode is triggered by:

- ML AD detection
- spike detection
- sustained loudness behaviour

AD Mode effects:

- immediate VOL_DOWN (1 step)
- block VOL_UP
- AD hold timer
- AD action limits

This is the core of AdBuster's anti-advertisement behaviour.

CEPA Logic PRO v13 — Context → Pattern → Decision → IR

1. Context Layers (Input)

- ML Context (AD / MUSIC / TALK / NORMAL)

- Perception Context (RMS / STD / DELTA / RANGE)
- Behavioural Context (CEPA State Machine)
- Time-of-Day Context
- User Context (Manual Override)
- Spike Context
- Drift & Trend Context
- Silence Context
- Safety Context (cooldowns, anti-spam, IR error block)

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2. Pattern Analysis (Interpretation)

- Trend Pattern
- Drift Pattern
- Spike Pattern
- Stability Pattern
- Deadzone Pattern
- Context Shift Pattern
- Return Pattern
- Human-like Timing Pattern
- Music Pattern
- Dialogue Pattern
- Advertisement Pattern

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3. CEPA Decision Logic

Inputs:

- level_smooth
- ml_flags
- patterns
- thresholds

Decisions:

- VOL_DOWN
- VOL_UP
- PASS

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4. IR Command Output

Commands:

- VOL_DOWN
- VOL_UP

Safety:

- cooldown timers
- anti-spam window
- AD block (no VOL_UP during ads)
- IR error protection

Summary

CEPA Logic PRO v13 processes audio through four layers:

1. Context
2. Pattern
3. Decision
4. IR Commands

This ensures stable, predictable, human-like volume behaviour.

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